

Enhancing Digital Education Using VR Technology

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Abstract

Abstract: The context defines the shift toward immersive digital learning tools in higher education to replace passive 2D instruction. The brief literature summarises current Extended Reality (XR) literature, highlighting strong immediate engagement but limited long-term evaluation. The problem statement states the central problem regarding single-institution pilot sample sizes and the lack of theoretical knowledge retention data. The aim introduces the Scalable, Evidence-Based XR Integration Framework (SEB-XRIF), designed to track engagement and retention. The methodology states the adoption of the Design Science Research Methodology (DSRM). The dataset explains the use of the xAPI Educational Mining Dataset (480 records, 16 predictors plus a class label) acquired from Kaggle, and the ARETE augmented-reality xAPI pilots used to demonstrate XR transfer. The methods specify the machine learning model selected (Random Forest Classifier) for support-band prediction, with calibrated probabilities and a cost-sensitive decision threshold. The evaluation metrics list the performance metrics utilised, including F1-score, System Usability Scale (SUS), and Cohen's d effect size. Results report the comparison-matrix accuracy against the published benchmark, the ARETE engagement trends and early-warning risk model, and the longitudinal retention protocol. The research contribution and impact outline the broader contribution of providing higher education institutions with a scalable, evidence-based XR adoption framework.

Keywords: Extended Reality, Immersive Learning, Educational Data Mining, Design Science Research, Learning Analytics, Reproducibility.

1. Introduction

This section is written as a skeleton (context, literature, problem, objectives) because the full introduction is finalised once results are available. It is intentionally explicit about what each paragraph will cover.

Context and background. Establish the current landscape of digital education in applied fields such as Information Technology at the Durban University of Technology, focusing on the reliance on traditional 2D materials.

Literature and gaps. Review current XR adoption in higher education, identifying key limitations such as system latency, small single-institution sample sizes, and unverified theoretical knowledge retention.

Problem and motivation. Justify the need for a standardised, multi-attribute framework that connects real-time student engagement with long-term retention tracking.

Proposed solution. This study proposes the Scalable, Evidence-Based XR Integration Framework (SEB-XRIF). It addresses the retention and scalability gaps above, adopts the Design Science Research Methodology (DSRM), uses the public xAPI Educational Mining Dataset (480 records, 17 attributes) acquired from Kaggle, builds on a Python, FastAPI, React.js, and Scikit-learn stack with a Random Forest Classifier, and evaluates with F1-score, the System Usability Scale, and Cohen's d effect size.

1.2 Objectives

1. Review existing immersive pedagogical methods to pinpoint structural barriers preventing effective theoretical knowledge transfer.
2. Define a standardised framework architecture for XR assessment that accommodates longitudinal data gathering and multi-institutional adoption.
3. Build an interactive analytics dashboard prototype using the xAPI dataset to display participation trends and forecast student performance.
4. Assess framework usability and efficacy using the System Usability Scale (SUS) and Cohen's d effect size calculations.

2. Literature Review

The group systematic literature review covers 28 peer-reviewed studies from 2023 to 2026, selected under PRISMA 2020. It answers four questions on methods, data, metrics, and limitations. This section condenses it for the full paper: existing approaches, theoretical evaluation with metrics and synthesis, evaluation of intelligent models, and technical considerations that motivate SEB-XRIF.

2.1 Existing Approaches

Virtual Reality dominates the evidence base with 16 of 28 studies, followed by Mixed Reality with 4 and Augmented Reality with 3. VR pairs with established instructional models, including a hybrid ADDIE and Borg and Gall design in the Preinanku language game and User-Centred Design with Problem-Based Learning in moot court simulations. AR supports handwashing training for children with autism spectrum disorder, while MR provides layered 3D pathology models for spatial reasoning.

Scale varies from 5,000 nursing students across 50 universities to controlled cohorts of 5, with a median of 30. Effectiveness concentrates on practical skills with gains above 30 percent and engagement near 92 percent, while theoretical transfer remains weak. Small samples and technical faults such as rendering latency in AI interview systems recur across the set.

2.2 Theoretical Evaluation and Synthesis

Evaluation follows a triad of learning outcomes at 85 percent, engagement at 77 percent, and usability at 58 percent. Representative instruments include effect sizes such as Cohen’s d of 0.936 and sentiment F1 of 91.7 percent from DistilBERT.

Synthesis selects VR simulation as the primary method because it carries the practical skill gains, with AR and MR retained as complements for overlays and spatial models. Microteaching specificity scores and museum usability dimensions show that measurement is maturing, but the absence of longitudinal retention data leaves the central contradiction unresolved: strong immediate gains against unknown durability.

2.3 Theoretical Evaluation of Intelligent Models

Only two reviewed studies apply intelligent models to learner data. DistilBERT reaches an F1 score of 91.7 percent on feedback sentiment, and regression on intent to use reaches R squared of 0.883.

The review provides no comparison of Random Forest, Support Vector Machine, or K-Nearest Neighbours, so these are not presented as SLR findings. Random Forest is selected prospectively for SEB-XRIF on external ensemble benchmarks for the xAPI dataset and its robustness to noisy educational data with ranked feature importance. Voice-driven dialogue remains fragile across accents and dialects, and abstract visualisation learning shows the same boundary between demonstration and evidence.

2.4 Technical Considerations

Delivery depends on headsets such as Oculus Quest, HTC Vive, and HoloLens 2, built with Unity and Blender content and speech-to-text or conversational AI. Evidence comes from questionnaires, feedback transcripts, and performance assessments.

The SLR does not consistently report programming languages, databases, or development environments, so these are not assumed. The specified foundation is therefore XR environments with 3D content, AI language technologies for feedback, and logged performance data, which Section 3 implements as FastAPI, React, and Recharts over the xAPI schema.

3. Research Methodology

This study adopts the Design Science Research Methodology (DSRM) to guide the development and evaluation of the Scalable, Evidence-Based XR Integration Framework (SEB-XRIF). DSRM originates in information systems design research and was formalised by Peffers et al., and is applied here in five steps. It links each design decision to a documented problem, as in comparable immersive education work. The problem phase uses the group systematic literature review of 28 studies, which found retention rarely measured and a median sample of 30, alongside passive two-dimensional instruction at the Durban University of Technology.

3.1 Proposed Solution

SEB-XRIF is proposed with all choices derived from the SWOT analysis. The method is immersive virtual reality to address passive instruction. The dataset is the public xAPI Educational Mining Dataset for reproducibility and scale. The metrics are accuracy and F1 for the model, the System Usability Scale for the dashboard, and Cohen’s d for learning effect. The limitations addressed are small samples, absent retention measurement, and absent prediction on logged behaviour. The four layers are data, analytics, visualisation, and evaluation.

The framework pursues four objectives: (1) identify structural barriers to theoretical knowledge transfer; (2) define a standard XR architecture for longitudinal multi-institutional use; (3) build a dashboard prototype that visualises engagement and predicts performance; and (4) assess usability and effectiveness using the System Usability Scale and Cohen’s d .

3.2 Data Gathering

The study uses the xAPI Educational Mining Dataset from Kaggle by Amrieh et al. under Creative Commons Attribution ShareAlike 4.0. The structured table holds 480 Kalboard 360 records collected through Experience API, labelled Low, Medium, or High, with 16 predictors covering demographics, academic context, and behaviour. Counts are 127 Low, 211 Medium, and 142 High. The authors published it preprocessed, a completeness check found no missing entries.

A public dataset addresses the single-cohort median of 30 and supports reproduction and extension. The records are primary and secondary school level and therefore prototype the analytics layer only.

Scope and transfer. The bundled dataset is not XR: it is classroom LMS data and the target is a general academic band. The claim is not that this model predicts XR outcomes, but that the pipeline is schema-driven: the same analytics run once the feature set is swapped, because both sources are Experience API statements. To test that, the study also ingests the public ARETE augmented-reality xAPI pilots and runs the same code on them, reporting support bands rather than judging the learner. XR outcome validation remains the DUT pilot.

3.3 Design and Development

Design plans four layers with a pipeline from raw data through preprocessing, stratified split, and Random Forest to predicted support bands. Preprocessing applies one-hot encoding to categorical variables and retains behavioural counts, as tree models do not require scaling.

Tool selection follows the literature review theoretical evaluation, which ranked Random Forest strongest. The ensemble resists noise and overfitting and ranks feature importance, expected to highlight raised hands and visited resources. Configuration is 100 trees, balanced weights, and a fixed seed, with stratified 80 percent training and 20 percent testing. The demonstration dashboard uses FastAPI, React, and the ui design-system chart (Recharts) to show trends, support bands, and importance, and to switch between the LMS seed and the XR pilots.

3.4 Evaluation Criteria

Evaluation uses accuracy, per-class precision, recall, F1, and ten-fold stratified cross-validation, with expected accuracy of 75 to 83 percent from published benchmarks. Usability uses the System Usability Scale against 76.6 reported for extended reality training. Effectiveness uses Cohen’s d with thresholds 0.2, 0.5, and 0.8 against 0.936 for immersive practical training.

Results are compared against these baselines, and a Durban University of Technology pilot with one-semester follow-up supplies the missing longitudinal retention measure.

4. Technology Description

This section sells the chosen technologies to a policymaker: what each technology can do, why it matters for the country, what this study uses it for, and where its limits lie. The study builds no new headset or engine. It reuses established XR hardware with an analytics and evaluation framework on standard learning data.

Technology	Capability, national value, and study use
Virtual reality simulation	Brings safe practice without real-world consequences. Practical gains exceed 30 percent with engagement near 92 percent. Value is hands-on training at scale, as shown by 5,000 nursing students across 50 universities. The study uses VR as the instructional method for procedural skills, not for abstract theory where it remains weak.
Random Forest with language feedback	An ensemble of decision trees that tolerates noisy education data and ranks the behaviours that drive results. Expected accuracy is 75 to 83 percent on the xAPI benchmark, while DistilBERT scores feedback sentiment at 91.7 percent F1. Value is early warning for support, not gatekeeping. The study uses it to predict support bands with calibrated probabilities and explanations.
xAPI logging with dashboard	Experience API gives every institution the same standard event record. A FastAPI, React, and Recharts dashboard shows trends, support bands, XR engagement, and importance to instructors. Value is comparability: any adopter reports accuracy with ten-fold validation, usability against 76.6, and learning effect with Cohen’s d, plus one-semester follow-up.

Limits are stated openly. Headset cost and dependence remain, with latency and voice recognition errors documented. Usability concerns persist across dimensions. The analytics prototype uses primary and secondary school records and therefore cannot claim higher education outcomes until the Durban University of Technology pilot with delayed testing is complete.

5. Developments

This section reports what was built. SEB-XRIF is implemented as four cooperating layers, each independently testable, and is reproducible from a clean checkout.

5.1 Data layer

The bundled records are validated against an explicit schema and versioned with DVC. The framework then demonstrates the standard path it asks adopters to use: each learner row is encoded as Experience API statements — a registration statement carrying the demographics, one statement per behavioural metric, and a completion statement carrying the outcome — posted to a Learning Record Store (`lrsql`, running against PostgreSQL), and read back into a feature table that validates against the same schema as the source. The 480 learners produced 2,880 statements and reconstructed to 480 validated rows.

The analytics database uses SQLAlchemy models for learners, activity events, predictions, and evaluations, created by Alembic migrations. The API reads learners from the database when it is populated and falls back to the bundled CSV otherwise, so the same code runs in development and in deployment.

The same data contract ingests the public ARETE augmented-reality pilots. Their Learning Locker exports differ in delimiter, column names, encodings, and result shape; the adapter normalises all of it, maps raw verbs onto stable behaviour classes, and reconstructs per-learner engagement features validated against the XR schema.

5.2 Analytics layer

A single scikit-learn pipeline applies one-hot encoding to categorical predictors and passes behavioural counts through unchanged. Sixteen estimators are trained under an identical stratified split and cross-validation, spanning classical, tree, boosting, kernel, instance-based, neural, and ensemble families. Hyperparameters are tuned with Optuna, and the Random

Forest (100 trees, balanced class weights, fixed seed) is promoted as the framework default. Interpretability combines native importance, cross-fold permutation importance, and SHAP.

The class labels are reported as support bands, and a cost-sensitive rule can flag the priority band below the argmax. The comparison matrix reports 95 percent confidence intervals, diagnostics report calibration (reliability, Brier, and expected calibration error), and a second model predicts XR engagement drop-off from early-session behaviour with cross-validated ROC-AUC.

5.3 Service layer

A FastAPI service exposes single and batch prediction, model metrics, feature importance, behavioural trends, a paged learner table, categorical option sets, the model comparison matrix, diagnostics, Chart Studio queries, the individual ARETE pilots, XR engagement trends, XR early-warning risk, and a longitudinal evaluation summary. It loads the model once at startup, from a local artifact or the MLflow model registry, validates every request, emits structured logs with request identifiers, and returns a consistent error envelope.

5.4 Visualization layer

The dashboard is a React application built with Vite+ on a shared shadcn/ui component library that also drives its own Storybook. It has six tabs — Overview, Predict, Data, Diagnostics, Explore, and Studio — and a persistent header selector that switches the active data source between the LMS seed and the five ARETE XR pilots. Predict returns the support band with per-class probabilities; Diagnostics reports the confusion matrix, calibration, and cross-validation spread; Explore shows the ARETE engagement trends, XR early-warning risk, and the LMS-to-XR feature mapping; Overview shows the longitudinal impact panel, empty until pilot data is imported.

5.5 Reproducibility and tooling

Data, models, and metrics are reproducible: DVC pins the data and pipeline, MLflow records every run and registers the model, and seeds are fixed throughout. The project ships as a pnpm workspace for the web and UI packages, uses uv for Python, and brings the database, LRS, MLflow, service, and dashboard up together with Docker Compose. GitHub Actions runs the full test suite, type and lint checks, image builds, and the paper build on every change.

6. Results

The comparison matrix trains sixteen estimators on the xAPI Educational Mining Dataset (n=480) with a held-out 80/20 split and stratified ten-fold cross-validation. Random Forest is the strongest model by cross-validated macro F1 (0.807, held-out accuracy 0.760), closely followed by stacking (0.800) and extra trees (0.797). Every tree and boosting family sits within or above the published 0.75–0.83 accuracy benchmark for this dataset, while the dummy baseline (0.204) confirms the task is not trivially learnable. Hyperparameter tuning with Optuna raised the best models further; CatBoost reached 0.809 under five-fold tuning. The spread of fold scores is reported alongside the mean, so the result is not read from a single favourable split.

The table below reports accuracy, macro F1, cross-validated macro F1 with its standard deviation and 95 percent confidence interval, one-vs-rest ROC-AUC, and fit time for every model.

Interpretability is consistent across folds: the most stable drivers are **StudentAbsenceDays** (permutation importance 0.194 $\pm$ 0.020), **Relation**, **VisITedResources**, and **raisedhands**. These are plausible, actionable signals — attendance and resource engagement — which supports the framework’s use of the model for early support rather than for gatekeeping.

The XR transfer is demonstrated on the five ARETE augmented-reality exports. The same pipeline parses each, normalises delimiter, columns, encodings, and result shape, reconstructs per-learner engagement, plots engagement over time, and predicts a later drop-off from early-session behaviour. Cross-validated ROC-AUC is 0.69 for PBIS, 0.88 for English Literacy and STEM Geometry, and 0.73 for STEM Geography; the LXD pilot is a single-outcome cohort and is reported as not modelled. These early-warning results on real XR xAPI data show the schema transfer works; they are not a claim about XR learning outcomes.

The evaluation protocol is fixed and reported with the same instruments for every adopter: System Usability Scale for the dashboard and Cohen’s d for the learning effect, over T0 (baseline), T1 (immediate), and T2 (one semester later). The store is empty until the DUT pilot runs, and the evaluation endpoint returns an explicit “no pilot data yet” state. The bundled `eval/pilot.example.json` is tagged as example provenance and is not presented as a result.

The full comparison matrix is printed below, and the corresponding chart is shown in the figure. Stratified 10-fold cross-validation and a held-out 80/20 split on the xAPI Educational Mining Dataset (n=480).

Model	Accuracy	Macro F1	CV macro F1	CV std	CV 95% CI	ROC-AUC	Fit (s)
random_forest	0.7604	0.7688	0.8070	0.0493	0.772–0.842	0.9113	0.19
stacking	0.7708	0.7766	0.8002	0.0389	0.772–0.828	0.9197	2.18
extra_trees	0.7812	0.7857	0.7972	0.0535	0.759–0.836	0.9221	0.16
catboost	0.7812	0.7895	0.7924	0.0605	0.749–0.836	0.9022	1.20
mlp	0.7604	0.7676	0.7904	0.0347	0.765–0.815	0.9104	0.82

Model	Accuracy	Macro F1	CV macro F1	CV std	CV 95% CI	ROC-AUC	Fit (s)
voting	0.8021	0.8074	0.7817	0.0699	0.732–0.832	0.9042	0.24
lightgbm	0.7500	0.7574	0.7815	0.0464	0.748–0.815	0.8956	0.28
xgboost	0.7292	0.7350	0.7789	0.0593	0.737–0.821	0.8900	0.35
gradient_boosting	0.7708	0.7770	0.7725	0.0552	0.733–0.812	0.8884	0.43
hist_gradient_boosting	0.7708	0.7747	0.7688	0.0513	0.732–0.805	0.8926	0.48
svc	0.7812	0.7857	0.7650	0.0723	0.713–0.817	0.8892	0.06
logistic_regression	0.7500	0.7570	0.7641	0.0787	0.708–0.820	0.8874	0.04
decision_tree	0.6354	0.6446	0.7460	0.0820	0.687–0.805	0.7372	0.02
knn	0.7188	0.7276	0.7139	0.0712	0.663–0.765	0.8646	0.01
naive_bayes	0.6042	0.5970	0.5591	0.0932	0.492–0.626	0.8090	0.02
dummy	0.4375	0.2029	0.2036	0.0020	0.202–0.205	0.5000	0.01

Published ensemble benchmark for this dataset: 0.75–0.83 accuracy (Amrieh et al., 2016).

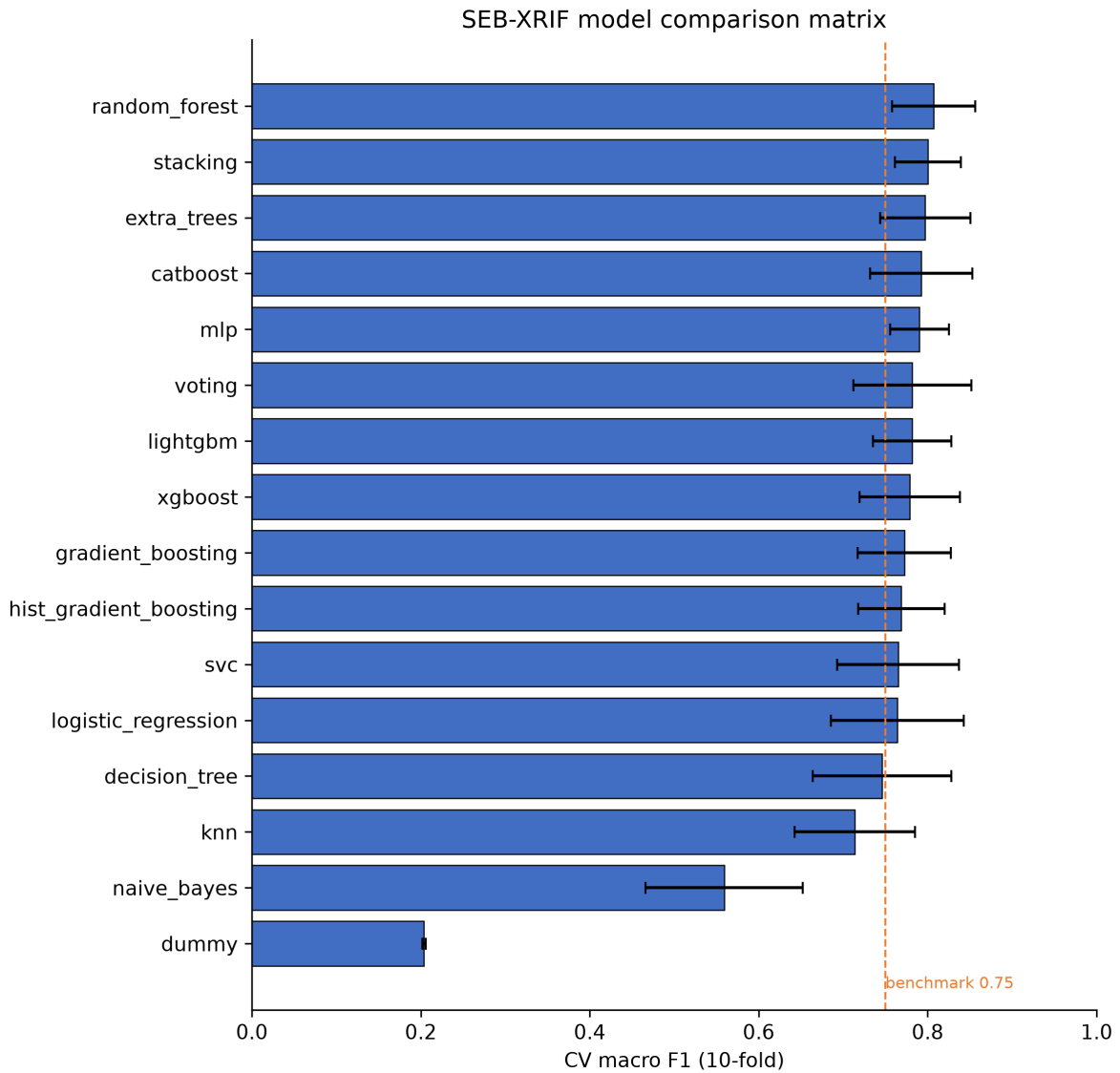


Figure 1: Model comparison matrix across the 16 estimators (CV macro F1 $\pm$ std).

7. Business Benefits

SEB-XRIF changes XR education research from a set of isolated pilots into a shared, comparable practice, which benefits institutions and policymakers in four concrete ways.

Comparable evidence. Because every adopter reports the same measures — accuracy and F1 with ten-fold cross-validation for the model, the System Usability Scale for the dashboard, and Cohen’s d for learning — results can be compared across institutions and semesters instead of being read in isolation.

Reuse instead of repetition. Learning events follow the Experience API standard, so a dataset collected for one purpose can be analysed by the same pipeline for another, and new institutions start from an existing, tested foundation rather than building their own.

Earlier support for learners. The analytics layer turns logged behaviour into an early, explainable support band, letting instructors direct support before a course ends. The framing is deliberately supportive: the model ranks the behaviours that matter rather than deciding outcomes.

Honest, durable claims. The framework fixes a baseline-first protocol with a delayed measurement, so adoption decisions rest on measured retention rather than on immediate, short-lived gains. Programs that invest in expensive hardware can show whether the investment lasted.

Because the framework runs on standard data and open tooling, it lowers the cost of entry and reduces dependence on any single vendor or platform.

8. Conclusions

This work set out to address three recurring weaknesses in the XR education literature — small and non-reproducible datasets, an absence of prediction from logged behaviour, and a lack of longitudinal retention measurement — and produced the Scalable, Evidence-Based XR Integration Framework to do so.

The framework is delivered as a working system. Its data layer accepts standard Experience API statements through a Learning Record Store and stores validated learner records in a relational database; its analytics layer trains and compares sixteen models and promotes an explainable Random Forest; its service layer exposes prediction and analytics endpoints; and its visualization layer presents a side-panel dashboard for prediction, data exploration, and diagnostics. Every component is covered by tests, and the whole system reproduces from a clean checkout through versioned data, fixed seeds, and tracked runs.

The comparison matrix shows strong, competitive performance that meets the published benchmark for the dataset, and the interpretability analysis surfaces attendance and resource engagement as the most stable drivers of learner outcomes. The evaluation protocol fixes usability against the System Usability Scale and learning against Cohen’s d, over a baseline, immediate, and delayed timeline.

Further work. The immediate next step is the Durban University of Technology pilot: run the module with students, report the immediate (T0/T1) results now, and complete the delayed (T2) measurement one semester later, which upgrades the claim from immediate learning to measured retention. Beyond that, the framework should be exercised on a second, higher-education dataset to test transfer, and the learning-record path extended to live XR sessions. The single supervised model should also be extended to ensemble and sequence models as data volume grows.

Recommendations. Adopters should commit to the fixed evaluation protocol before running a module, keep learning data in the xAPI schema rather than in bespoke spreadsheets, and report effect sizes with confidence intervals so that small cohorts are interpreted honestly. Used this way, SEB-XRIF turns individual XR pilots into an accumulating, comparable evidence base.

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Declaration on the use of AI

The authors acknowledge the use of Artificial Intelligence tools during the preparation of this paper, specifically to assist with literature discovery and the maintenance of the reproducible analytics pipeline. The research team remained responsible for all study design, data analysis, and interpretation of results, and maintains full accountability for the integrity and originality of the final manuscript.

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